

The goal of this lecture is to prove the following beautiful theorem, which was one of the first results in the field of *discrepancy theory*, which studies how “evenly distributed” sequences are. Buckle your seatbelts. This proof is going to be a hike, but I promise the view is worth it.

Theorem 1 (Roth 1954, [On irregularities of distribution](#)). *There exists a constant $c > 0$ such that the following condition holds: for all sequences $(P_i)_{1 \leq i \leq N}$ with $P_i \in [0, 1)$ for all i , there exists n with $1 \leq n \leq N$ and $x \in [0, 1)$ such that*

$$|\#\{i \mid i \leq n, P_i < x\} - nx| > c\sqrt{\log N}.$$

There are four versions of [Theorem 1](#). Here they are in order of increasing difficulty.

- Replace $\sqrt{\log N}$ with 1. This one is trivial.
- Replace $\sqrt{\log N}$ with $\frac{\log \log N}{\log \log \log N}$. This one was proven by van Aardenne-Ehrenfest in 1945.
- The original [Theorem 1](#), proven by Roth in 1954.
- Replace $\sqrt{\log N}$ with $\log N$. This was proven by Schmidt in 1972. This is the best possible.

We will prove [Theorem 1](#). We claim the following lemma, which finishes the proof when applied to the sequence $(P_i, \frac{i-1}{N})_{1 \leq i \leq N}$.

Lemma 2. *There exists a constant $c > 0$ such that the following condition holds: for all sequences $(P_i)_{1 \leq i \leq N}$ with $P_i \in [0, 1)^2$ for all i , there exists $(x_1, x_2) \in [0, 1)^2$ such that*

$$|\#\{i \mid P_i \in [0, x_1) \times [0, x_2)\} - Nx_1x_2| > c\sqrt{\log N}.$$

Let $c = c_1$ be some constant to be determined later. We begin with some definitions. For $x = (x_1, x_2) \in [0, 1)^2$, let

$$C(x) = \#\{i \mid P_i \in [0, x_1) \times [0, x_2)\} \quad \text{and} \quad D(x) = C(x) - Nx_1x_2.$$

Hence we need to show $|D(x)| > c_1\sqrt{\log N}$ for some x .

Let $R = [0, 1)^2$. A *rectangle* is a subset of R of the form $[a, b) \times [c, d)$ for some real a, b, c, d with $a < b$ and $c < d$. For a rectangle $r = [a, b) \times [c, d)$, we define the four sub-rectangles

$$\begin{aligned} r_{01} &= [a, \frac{a+b}{2}) \times [\frac{c+d}{2}, d) & r_{11} &= [\frac{a+b}{2}, b) \times [\frac{c+d}{2}, d) \\ r_{00} &= [a, \frac{a+b}{2}) \times [c, \frac{c+d}{2}) & r_{10} &= [\frac{a+b}{2}, b) \times [c, \frac{c+d}{2}) \end{aligned}$$

In other words, let $r_{00}, r_{10}, r_{01}, r_{11}$ denote its bottom-left, bottom-right, top-left, and top-right quadrants, respectively. Additionally, we say a rectangle is *bad* if it contains an element of P , and *good* otherwise.

Let $c_2 = c_1^2$. It's enough to show

$$\int_R D(x)^2 dx > c_2 \log N,$$

since if $|D(x)|$ was always less than $c_1\sqrt{\log N}$, then the integral would be less than $c_2 \log N$. We proceed by bounding this integral.

Let m be the integer such that $2N \leq 2^m < 4N$. For all i with $0 \leq i \leq m$, let $\mathcal{P}_i$ be the partition of R formed by the 2^m rectangles

$$\left[\frac{a}{2^i}, \frac{a+1}{2^i} \right) \times \left[\frac{b}{2^{m-i}}, \frac{b+1}{2^{m-i}} \right) \quad 0 \leq a < 2^i, 0 \leq b < 2^{m-i}.$$

In other words, $\mathcal{P}_i$ divides the unit square into a grid of 2^m rectangles, each of them 2^{-i} by 2^{i-m} . For any i and x , let $r_i(x)$ be the unique rectangle such that $x \in r_i(x)$ and $r_i(x) \in \mathcal{P}_i$. Then define

$$f_i(x) = \begin{cases} 0 & \text{if } r_i(x) \text{ is bad,} \\ 1 & \text{if } r_i(x) \text{ is good and } x \in r_i(x)_{00} \text{ or } x \in r_i(x)_{11} \\ -1 & \text{if } r_i(x) \text{ is good and } x \in r_i(x)_{01} \text{ or } x \in r_i(x)_{10} \end{cases}$$

and $F(x) = f_0(x) + \cdots + f_m(x)$.

From Cauchy-Schwarz on L^2 , we see

$$\int_R D(x)^2 dx \geq \frac{(\int_R D(x)F(x) dx)^2}{\int_R F(x)^2 dx}.$$

We now bound the numerator and denominator of the right side.

For the denominator, observe that for all i , $\int_R f_i(x)^2 dx < 1$, and for any $i \neq j$, $\int_R f_i(x)f_j(x) dx = 0$. Therefore,

$$\int_R F(x)^2 dx = \int_R f_0(x)^2 + \cdots + f_m(x)^2 dx < \int_R (m+1) dx = m+1.$$

For the numerator, we split the integral into f_i s and good rectangles r . For a good rectangle r , let a and b be the vectors from the bottom-left corner to halfway along the bottom and left edges, respectively. We compute

$$\begin{aligned} \int_r D(x)f_i(x) dx &= \int_{r_{00}} D(x)f_i(x) dx + \int_{r_{01}} D(x)f_i(x) dx \\ &\quad + \int_{r_{10}} D(x)f_i(x) dx + \int_{r_{11}} D(x)f_i(x) dx \\ &= \int_{r_{00}} (D(x)f_i(x) + D(x+a)f_i(x+a) \\ &\quad + D(x+b)f_i(x+b) + D(x+a+b)f_i(x+a+b)) dx \\ &= \int_{r_{00}} (D(x) - D(x+a) - D(x+b) + D(x+a+b)) dx \\ &= \int_{r_{00}} (C(x) - C(x+a) - C(x+b) + C(x+a+b) - N|a||b|) dx \\ &= -N|a|^2|b|^2 = -\frac{N}{2^{2m}}. \end{aligned}$$

By pigeonhole, there are at least N good rectangles in each $\mathcal{P}_i$, so we bound

$$\begin{aligned} \left(\int_R D(x)F(x) dx \right)^2 &= \left(\sum_{i=0}^m \sum_{\text{good } r \in \mathcal{P}_i} \int_r D(x)f_i(x) dx \right)^2 \\ &= \left(\sum_{i=0}^m \sum_{\text{good } r \in \mathcal{P}_i} \frac{N}{2^{2m}} \right)^2 \end{aligned}$$

$$\geq \frac{N^4(m+1)^2}{2^{4m}}.$$

Combining the bounds, we obtain

$$\int_R D(x)^2 \, dx \geq \frac{(\int_R D(x)F(x) \, dx)^2}{\int_R F(x)^2 \, dx} \geq \frac{N^4(m+1)}{2^{4m}} \geq \frac{1}{2^7} + \frac{\log N}{2^8 \log 2},$$

as desired.